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CL-1 Lab Assignment 5 (Group B): Data Aggregation

Analyzing Sales Performance by Region in a Retail Company

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (10, 5)
plt.rcParams["axes.grid"] = True
```

```
FILE = "Retail_Sales_Data.csv"

try:
    df = pd.read_csv(FILE)
except FileNotFoundError:
    print("Retail_Sales_Data.csv was not found!")
else:
    print(f"Loaded {FILE}")

display(df.head())
```

```
Loaded Retail_Sales_Data.csv
```

	Transaction_Date	Region	Product_Category	Quantity	Sales_Amount
0	2025-01-09	South	Grocery	1	2439.78
1	2025-03-11	West	Furniture	1	3831.02
2	2025-02-28	North	Grocery	2	4526.31
3	2025-02-09	West	Electronics	8	3631.41
4	2025-02-08	East	Furniture	5	4819.73

```
print("Shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())
print("\nData types:")
display(df.dtypes.to_frame("dtype"))
print("\nMissing values:")
display(df.isna().sum().to_frame("missing"))
print("\nSummary statistics:")
display(df.describe(include="all").T)
```

```
Shape: (300, 5)

Columns:
['Transaction_Date', 'Region', 'Product_Category', 'Quantity', 'Sales_Amount']

Data types:
```

	dtype
Transaction_Date	object
Region	object
Product_Category	object
Quantity	int64
Sales_Amount	float64

```
Missing values:
```

	missing
Transaction_Date	0
Region	0
Product_Category	0
Quantity	0
Sales_Amount	0

```
Summary statistics:
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Transaction_Date	300	87	2025-03-12	10	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Region	300	4	West	85	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Product_Category	300	4	Grocery	85	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Quantity	300.0	NaN	NaN	NaN	5.02	2.441202	1.0	3.0	5.0	7.0	9.0
Sales_Amount	300.0	NaN	NaN	NaN	2536.277933	1461.812002	106.04	1278.1725	2560.31	3798.195	4993.84

```python
# Identify relevant variables using common column names.
def find_col(columns, keywords):
    for c in columns:
        x = c.lower().replace(" ", "_")
        if any(k in x for k in keywords):
            return c
    return None

region_col = find_col(df.columns, ["region"])
sales_col = find_col(df.columns, ["sales_amount", "sales", "amount", "revenue"])
category_col = find_col(df.columns, ["product_category", "category", "product"])

print("Region column:", region_col)
print("Sales amount column:", sales_col)
print("Product category column:", category_col)

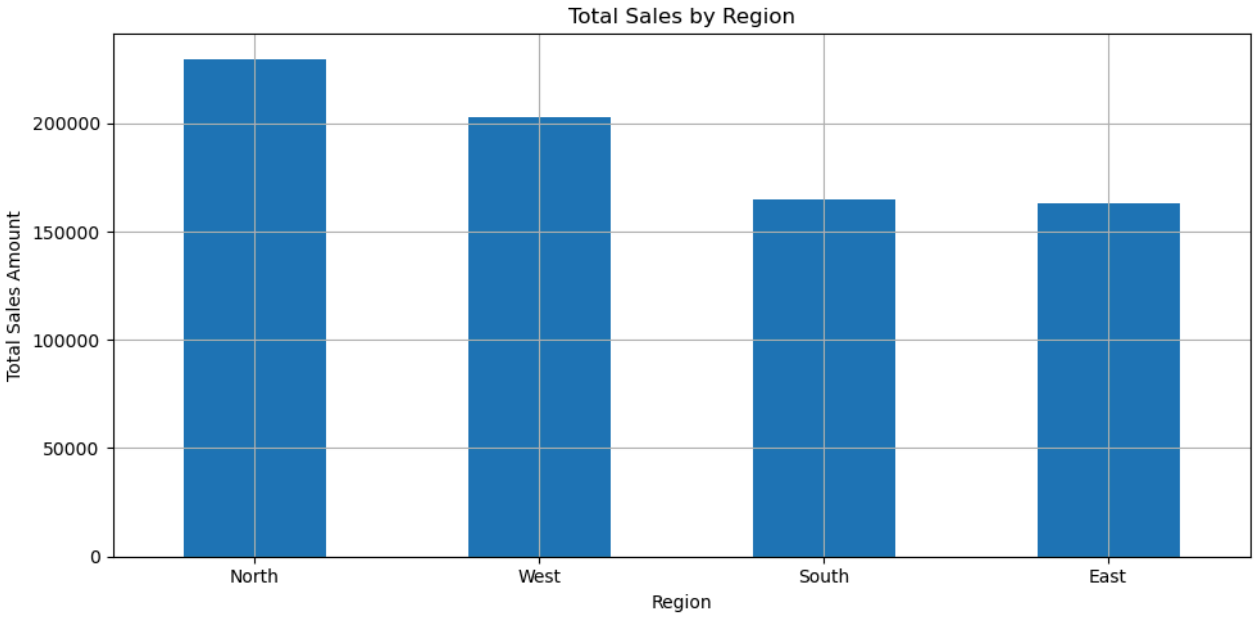
if sales_col:
    df[sales_col] = pd.to_numeric(df[sales_col], errors="coerce")
```

```
Region column: Region
Sales amount column: Sales_Amount
Product category column: Product_Category
```

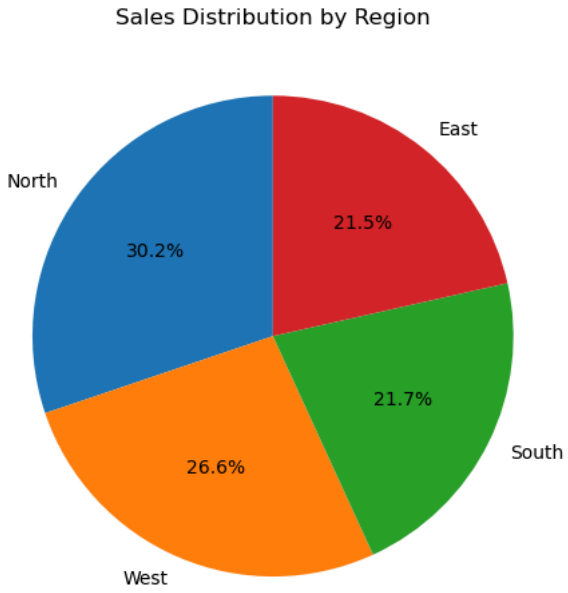
```python
if region_col and sales_col:
    region_sales = df.groupby(region_col)[sales_col].sum().sort_values(ascending=False)
    display(region_sales.to_frame("Total Sales"))

    plt.figure()
    region_sales.plot(kind="bar")
    plt.title("Total Sales by Region")
    plt.xlabel("Region")
    plt.ylabel("Total Sales Amount")
    plt.xticks(rotation=0)
    plt.tight_layout()
    plt.show()
else:
    print("Required columns were not identified.")
```

Region	Total Sales
North	229750.53
West	202726.49
South	165114.59
East	163291.77




```
if region_col and sales_col:
    region_sales = df.groupby(region_col)[sales_col].sum().sort_values(ascending=False)
    plt.figure()
    region_sales.plot(kind="pie", autopct="%1.1f%%", startangle=90)
    plt.title("Sales Distribution by Region")
    plt.ylabel("")
    plt.tight_layout()
    plt.show()
```



```
if region_col and sales_col:
    region_sales = df.groupby(region_col)[sales_col].sum().sort_values(ascending=False)
    top_regions = region_sales.head()
    print("Top-performing regions based on total sales:")
    display(top_regions.to_frame("Total Sales"))

    if len(region_sales):
        print(f"Highest-sales region: {region_sales.index[0]}")
```

```
Top-performing regions based on total sales:
```

Region	Total Sales
North	229750.53
West	202726.49
South	165114.59
East	163291.77

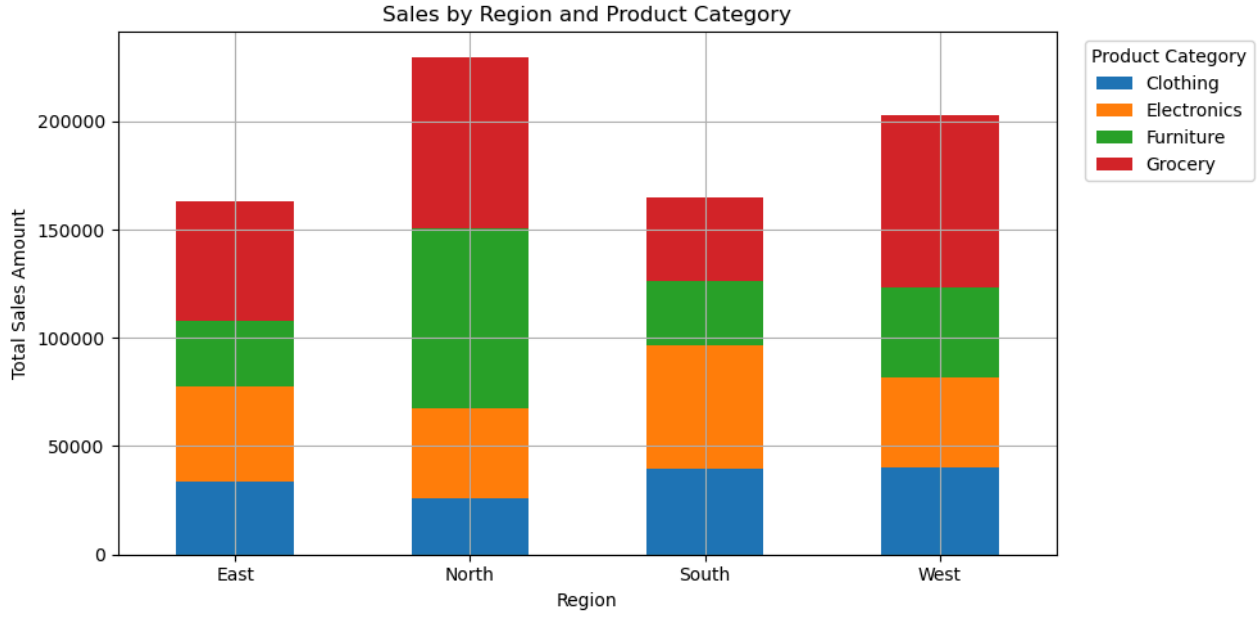
```
Highest-sales region: North
```

```
if region_col and category_col and sales_col:
    region_category_sales = (
        df.groupby([region_col, category_col])[sales_col]
          .sum()
          .unstack(fill_value=0)
    )
    display(region_category_sales)
```

Region	Product_Category Clothing	Electronics	Furniture	Grocery
East	33689.48	43755.41	30159.84	55687.04
North	25853.89	41592.80	83013.63	79290.21
South	39633.37	56684.34	30164.85	38632.03
West	40304.56	41444.73	41504.48	79472.72

```
if region_col and category_col and sales_col:
    plt.figure()
    region_category_sales.plot(kind="bar", stacked=True, figsize=(10, 5))
    plt.title("Sales by Region and Product Category")
    plt.xlabel("Region")
    plt.ylabel("Total Sales Amount")
    plt.xticks(rotation=0)
    plt.legend(title="Product Category", bbox_to_anchor=(1.02, 1), loc="upper left")
    plt.tight_layout()
    plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```



```
if region_col and category_col and sales_col:
    plt.figure()
    region_category_sales.plot(kind="bar", stacked=False, figsize=(10, 5))
    plt.title("Grouped Sales Comparison by Region and Product Category")
    plt.xlabel("Region")
    plt.ylabel("Total Sales Amount")
    plt.xticks(rotation=0)
    plt.legend(title="Product Category", bbox_to_anchor=(1.02, 1), loc="upper left")
    plt.tight_layout()
    plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```

